

EARLTON, ON, Canada

WMO: 716840

Lat: 47.6976N

Lon: 79.8535W

Elev: 800

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-18.4	-31.4	1.0	-23.5	-25.5	1.4	-18.2	25.5	17.2	23.4	14.7	5.6	310	0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	85.1	69.4	81.7	67.3	78.5	65.3	71.8	81.0	69.9	77.7	68.0	75.0	12.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.0	109.7	76.2	67.2	102.8	74.1	65.4	96.5	72.1	36.1	81.5	34.4	77.7	32.8	75.2	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	20.7	18.5	-33.4	90.8	6.9	3.2	-38.3	93.1	-42.3	94.9	-46.2	96.7	-51.2	99.0		
(5)			-33.6	75.3	6.8	2.2	-38.5	76.8	-42.5	78.1	-46.3	79.3	-51.3	80.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	6.0	8.4	20.2	35.7	51.6	60.7	65.0	63.0	55.4	42.6	29.2	14.1		
	DBStd	23.21	14.67	12.18	12.33	9.66	8.79	7.19	5.88	6.06	8.20	8.48	10.35	13.21		
	HDD50	6021	1364	1166	924	436	88	5	0	0	36	264	626	1112		
	HDD65	10104	1829	1586	1388	878	424	167	72	111	304	694	1074	1577		
	CDD50	1576	0	0	1	8	137	327	466	403	198	35	1	0		
	CDD65	185	0	0	0	0	8	38	73	49	16	1	0	0		
	CDH74	1919	0	0	1	5	127	441	736	454	148	7	0	0		
	CDH80	479	0	0	0	0	29	115	195	112	27	1	0	0		
	WSAvg	8.7	8.9	9.2	9.3	9.8	9.4	8.1	7.5	7.3	8.1	9.0	9.3	9.0		
	PrecAvg	29.7	1.8	1.5	1.7	1.9	2.7	3.1	3.7	2.8	3.4	3.0	2.4	1.8		
	PrecMax	35.2	3.5	4.2	3.8	3.0	5.4	4.6	6.4	4.9	6.6	4.5	3.8	3.1		
(15) Precipitation	PrecMin	25.6	0.9	0.4	0.1	0.6	1.4	1.7	1.7	0.8	1.0	0.8	1.0	0.9		
	PrecStd	2.6	0.7	0.9	1.1	0.6	1.2	0.9	1.2	1.1	1.2	1.2	0.7	0.6		
	DB	38.8	39.4	56.2	70.9	84.2	88.0	89.7	88.0	83.3	72.6	57.1	42.8			
	MCWB	37.4	36.4	50.6	56.5	66.0	70.5	72.4	71.2	69.9	61.9	51.5	39.9			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	35.1	34.6	45.3	62.4	77.0	83.2	85.0	83.1	78.2	66.6	50.8	37.6			
	2% MCWB	34.1	32.7	39.8	50.3	59.7	67.4	69.5	68.9	66.4	58.5	47.3	36.1			
	5% DB	32.6	31.1	41.1	56.3	72.7	79.2	81.5	79.0	74.2	61.5	47.1	34.7			
	5% MCWB	31.5	29.5	36.4	46.7	58.9	64.5	67.5	66.3	64.2	55.6	44.3	33.5			
	10% DB	28.8	27.5	37.2	50.4	68.1	75.0	78.1	75.3	70.1	56.5	43.1	32.4			
	10% MCWB	27.7	26.0	33.8	42.9	56.5	62.8	65.9	65.4	63.1	52.3	40.8	31.5			
	0.4% WB	37.8	36.6	51.9	58.8	69.7	73.5	75.5	73.4	71.5	64.1	52.8	40.9			
	0.4% MCDB	38.5	38.6	55.1	69.2	77.8	84.2	85.6	83.3	79.5	71.6	56.1	41.9			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	34.3	32.8	40.5	51.7	64.5	70.2	72.0	70.8	68.9	59.7	47.9	36.3			
	2% MCDB	34.9	34.3	44.0	59.3	72.5	79.3	80.4	79.7	75.3	64.9	49.8	37.2			
	5% WB	31.5	29.6	36.8	47.7	61.1	67.4	69.8	68.7	66.6	56.4	44.5	33.6			
	5% MCDB	32.5	30.9	40.1	54.7	69.0	75.2	77.2	75.9	72.1	60.7	46.8	34.6			
	10% WB	27.7	26.0	34.0	43.7	57.4	64.7	67.7	66.7	63.6	52.6	40.9	31.7			
	10% MCDB	29.0	27.4	36.7	49.9	65.8	72.0	75.3	73.0	69.3	56.3	43.0	32.5			
	MDBR	18.6	20.0	20.5	18.8	22.4	21.5	21.2	20.4	20.2	15.3	12.9	15.2			
	5% DB	14.3	18.8	21.5	29.6	29.2	27.3	26.2	25.9	25.7	23.8	18.0	13.0			
(36) Mean Daily Temperature Range	5% DB	13.9	17.5	16.6	19.1	15.8	14.5	13.2	13.5	15.0	16.4	15.0	12.4			
	5% WB	14.2	18.4	18.7	26.9	24.7	22.6	22.1	21.9	21.2	22.1	16.8	12.2			
	MCWBR	13.9	17.3	15.2	19.1	15.1	13.7	12.7	12.7	13.6	16.5	14.7	12.1			
	MCWBR	13.9	17.3	15.2	19.1	15.1	13.7	12.7	12.7	13.6	16.5	14.7	12.1			
(40) Clear-Sky Solar Irradiance	taub	0.265	0.282	0.314	0.348	0.363	0.385	0.397	0.386	0.364	0.329	0.297	0.264			
	taud	2.283	2.225	2.244	2.344	2.397	2.355	2.350	2.373	2.417	2.502	2.481	2.362			
	Ebn at Noon	259	279	286	287	285	279	273	272	267	261	249	246			
	Edh at Noon	23	31	36	36	36	38	38	35	31	24	20	19			
(44) All-Sky Solar Radiation	RadAvg	394	728	1199	1490	1664	1788	1752	1470	1113	658	348	279			
	RadStd	62	108	128	130	137	120	142	99	106	87	38	30			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-1.31	-1.30	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page